

AML — Secondary & Therapy-Related Origins



1. Origins Dock

WHAT YOU SEE

'Scene 42 — The Origins Dock' title.

WHAT IT ENCODES

This scene covers how AML arises: de novo, secondary to a prior myeloid disorder (MDS/MPN), therapy-related (chemo/radiation), and germline predisposition. Secondary and therapy-related AML carry worse prognosis.

BOARD TRAP

Secondary/therapy-related AML is typically adverse-risk (often TP53, complex karyotype) and frequently warrants transplant — distinguish from de novo AML.

2. Therapy-associated ships

WHAT YOU SEE

'THERAPY-ASSOCIATED SHIPS' banner over warships.

WHAT IT ENCODES

Therapy-related AML (t-AML) follows prior cytotoxic therapy and is an adverse, often TP53-mutated, MDS-related entity with poor response to standard chemo.

BOARD TRAP

t-AML's poor outcome stems from its biology (TP53, complex karyotype), not merely from prior treatment burden — molecular features drive prognosis.

3. Alkylating ship 5-7 yr

WHAT YOU SEE

'-5 -7' flag, clock '4-6 YR', 'ALKYLATING / CYCLO / BUSU / MELPH'.

WHAT IT ENCODES

Alkylator-related AML appears ~5–7 years after exposure (cyclophosphamide, busulfan, melphalan), preceded by MDS, with deletions of chromosome 5 and 7 (–5/del5q, –7/del7q) and complex karyotype.

BOARD TRAP

Alkylator t-AML: long latency (5–7 yr), antecedent MDS, –5/–7 — opposite profile to topoisomerase II–related t-AML.

4. –5/–7 flag

WHAT YOU SEE

Skull flag '–5 –7'.

WHAT IT ENCODES

Monosomy 5/7 and complex karyotypes are hallmarks of alkylator-related and MDS-related AML, both adverse-risk.

BOARD TRAP

–7/del(7q) and –5/del(5q) are adverse cytogenetics; don't confuse del(5q) here with the favorable isolated del(5q) of MDS (which responds to lenalidomide).

5. Topo II ship 1-3 yr

WHAT YOU SEE

Clock '1-3 YR', 'ETOP', 'TOPO II', 'ANTHRA'.

WHAT IT ENCODES

Topoisomerase II inhibitor–related AML (etoposide, anthracyclines) has SHORT latency (~1–3 years), no preceding MDS, and balanced translocations involving KMT2A (11q23) — and t(15;17) APL.

BOARD TRAP

Topo II t-AML: short latency, KMT2A/11q23 translocation, NO antecedent MDS — the inverse of the alkylator pattern. This is the high-yield contrast.

6. KMT2A 11q23

WHAT YOU SEE

'KMT2A 11q23' flag on the topo II ship.

WHAT IT ENCODES

KMT2A (MLL) rearrangements at 11q23 characterize topoisomerase II–related AML, often monocytic morphology. Most are adverse risk.

BOARD TRAP

11q23/KMT2A rearrangements are generally adverse; only t(9;11) KMT2A is intermediate. Latency and chromosome (11q23 vs –5/–7) tell you the causative drug class.

7. Chip watchtower / epigenetic

WHAT YOU SEE

'CHIP WATCHTOWER', paintbrush scrolls, 'DNMT3A / TET2 / ASXL1'.

WHAT IT ENCODES

Clonal hematopoiesis (CHIP) from age-related mutations in epigenetic regulators DNMT3A, TET2, ASXL1 raises AML risk. These 'painters over epigenetic regulation' precede overt leukemia.

BOARD TRAP

CHIP mutations (DNMT3A/TET2/ASXL1) are pre-leukemic and common with aging; they confer risk but are not diagnostic of AML by themselves.

8. 10x risk

WHAT YOU SEE

'10x' multiplier on the watchtower.

WHAT IT ENCODES

CHIP and antecedent MDS markedly increase the risk of progression to AML compared with baseline, reflecting a multistep clonal evolution.

BOARD TRAP

CHIP raises relative risk substantially but the absolute annual transformation rate stays low (~0.5–1%/yr) — don't treat asymptomatic CHIP as leukemia.

9. Second hit / multilineage dysplasia

WHAT YOU SEE

'SECOND HIT' lightning, 'multilineage dysplastic cells'.

WHAT IT ENCODES

AML-MR (myelodysplasia-related) arises from antecedent MDS/MPN or carries MDS-defining cytogenetics/mutations, with multilineage dysplasia. Adverse biology requiring a 'second hit'.

BOARD TRAP

Multilineage dysplasia and MDS-related cytogenetics define a distinct adverse AML subtype (AML-MR), separate from de novo AML even at the same blast count.

10. Monocytic clogging

WHAT YOU SEE

'MONOCYTE-SHAPED CREATURES' clogging pipes with yellow plaque.

WHAT IT ENCODES

Monocytic/myelomonocytic AML (and KMT2A-rearranged disease) shows monocytoid blasts, gum infiltration, and tissue involvement; risk of leukostasis when WBC is very high.

BOARD TRAP

Monocytic AML predisposes to extramedullary disease (gingival hyperplasia, skin) and leukostasis — hyperleukocytosis is an emergency needing cytoreduction.

11. Radiation symbol

WHAT YOU SEE

Yellow 'RADIATION SYMBOL' buoy, 'COMBINED = HIGHEST RISK'.

WHAT IT ENCODES

Ionizing radiation is a leukemogen; combined chemo + radiation gives the highest therapy-related AML risk. Reinforces the multifactorial origin theme.

BOARD TRAP

Combined modality therapy carries higher t-AML risk than either alone — relevant when counseling cancer survivors.

12. Benzene / chemical barrels

WHAT YOU SEE

'EXPOSURE BERTH' barrels 'BENZENE / CHLOR'.

WHAT IT ENCODES

Environmental/occupational benzene exposure is an established AML risk factor, acting through marrow toxicity and clonal selection.

BOARD TRAP

Benzene (and prior chemo/radiation) is an exogenous cause; germline predisposition (next) is endogenous — boards test recognizing the exposure history.

13. Germline pier

WHAT YOU SEE

'GERMLINE PIER' with figures and a lantern.

WHAT IT ENCODES

Germline predisposition (e.g. RUNX1, CEBPA, GATA2, DDX41, Fanconi anemia) causes familial AML/MDS. Suspect with young age, family history, or affected potential donors.

BOARD TRAP

Germline RUNX1/CEBPA/DDX41 mandates screening family members BEFORE using them as transplant donors — a related donor could carry the same predisposing mutation.
